UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIŚ 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

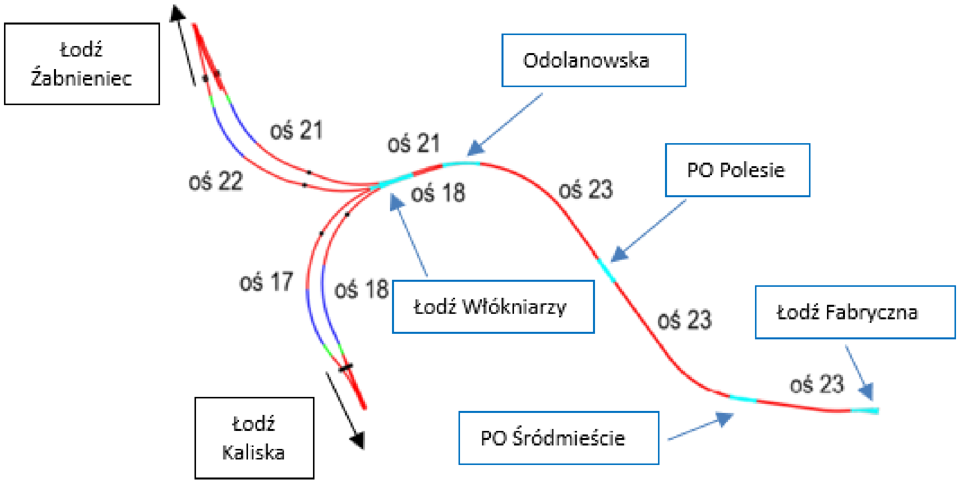

Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 06/09/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+628.92 after the erection of ring n. 269; which corresponds to a total excavation length of 337.55 m from the starting chainage.

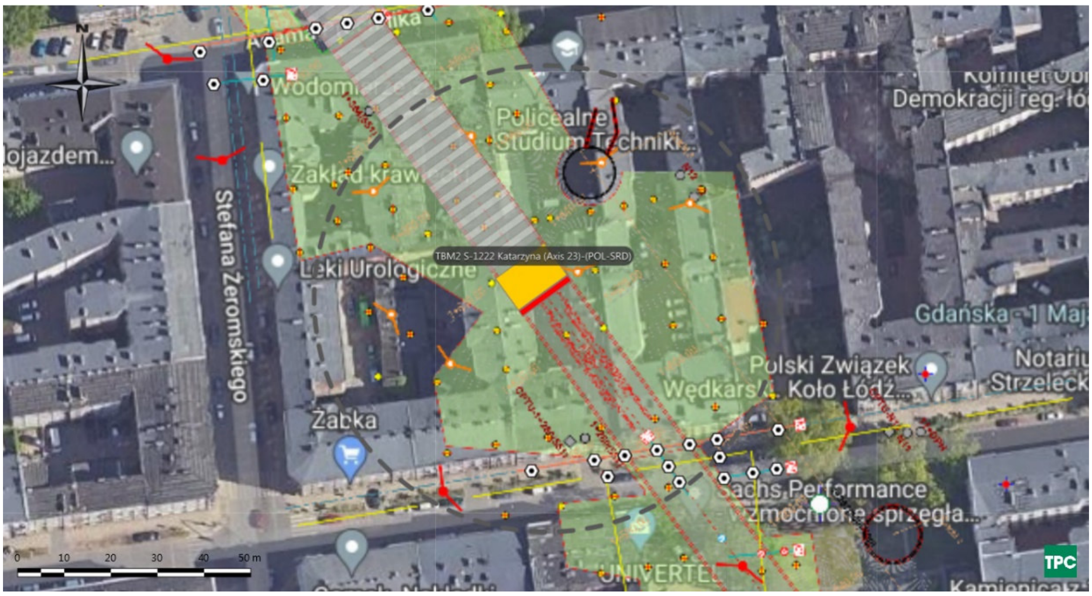

Figure 1-2 Plan view of TBM Position on 09/09/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown during last three days is at about 1.7 bar.

The pressure at the crown was maintained at an average value of about 1.6 bar on 06/09/2024, i.e. the attention limit. After the excavation stoppage, from 07/09/2024, the pressure at the crown increased to 1.8 bar, the recommended pressure in the project, remaining constant.

Figure 2-1 shows EPB sensors.

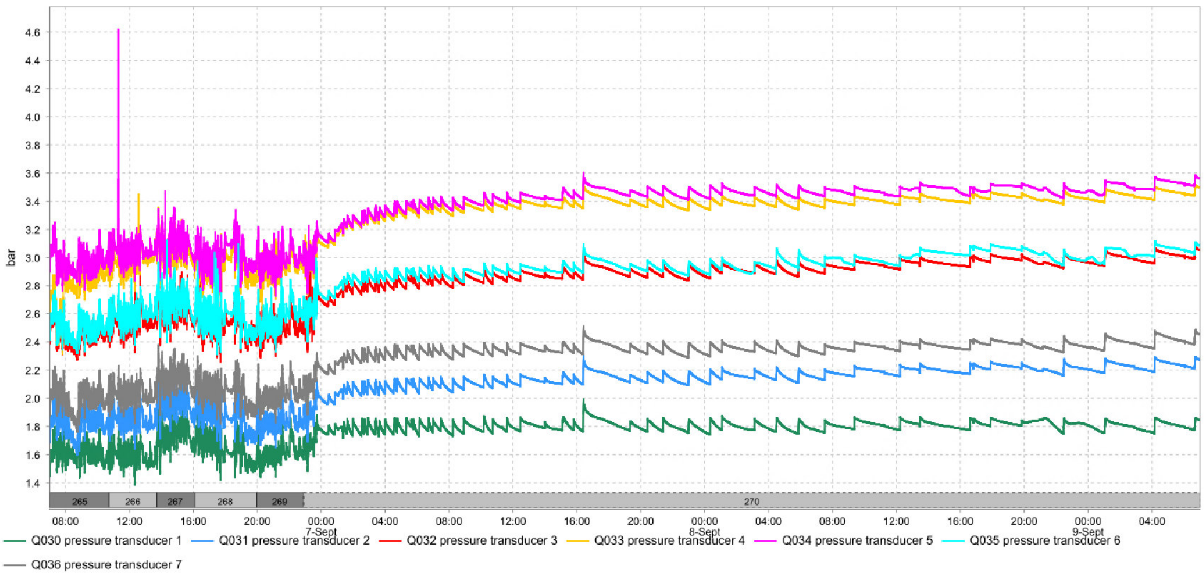

Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.25 bar indicating that the bulk chamber is correctly filled with material. Delta value has shown a drop during the excavation of ring 265, indicating partial void in the upper part of the chamber. After the excavation stoppage, from 07/09/2024, the delta pressure between sensors 1 and 2 increased to 0.4 bar, object of further investigations.

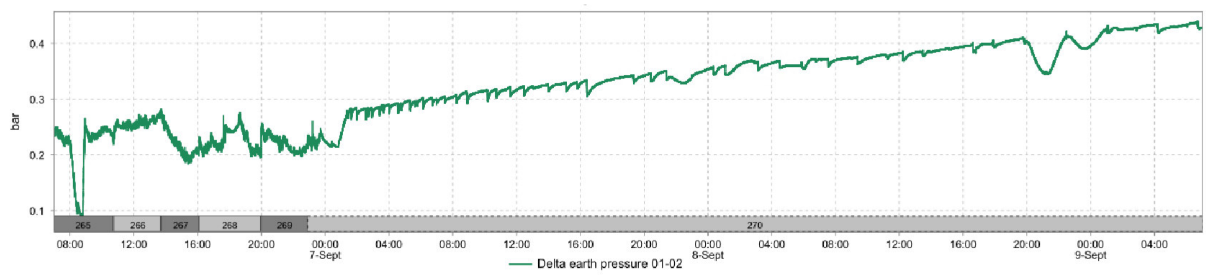

Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values lower than n. 1. Excavated material weight is higher than expected (450 t) for ring 265 and 268. The belt scale value is a raw indication of the excavated material quantity.

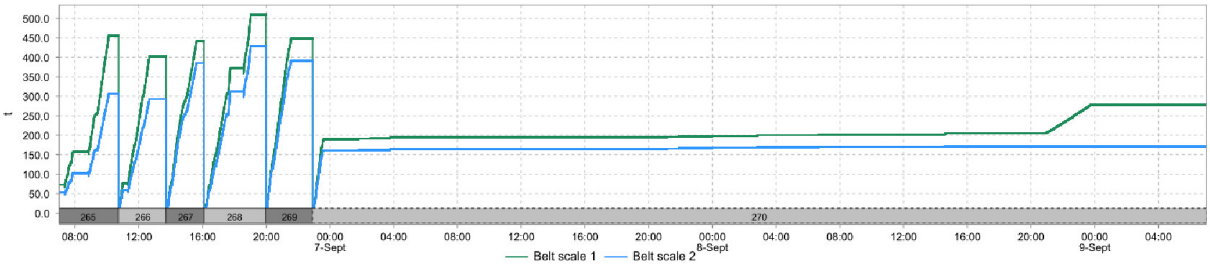

Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
265	456	307
266	402	292
267	441	385
268	509	429
269	449	391

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 8.6 MNm.

During advancement, average advance rate is of 15 mm/min.

During excavation, thrust force shows an average value of 65.8 MN and a maximum of 74 MN.

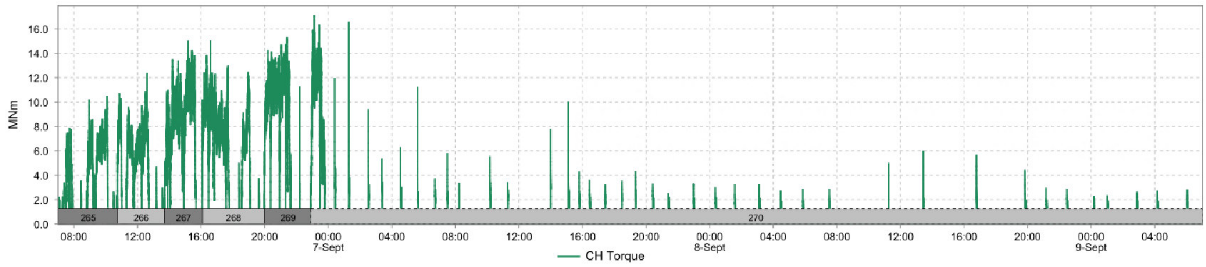

Figure 3-1 CH Torque

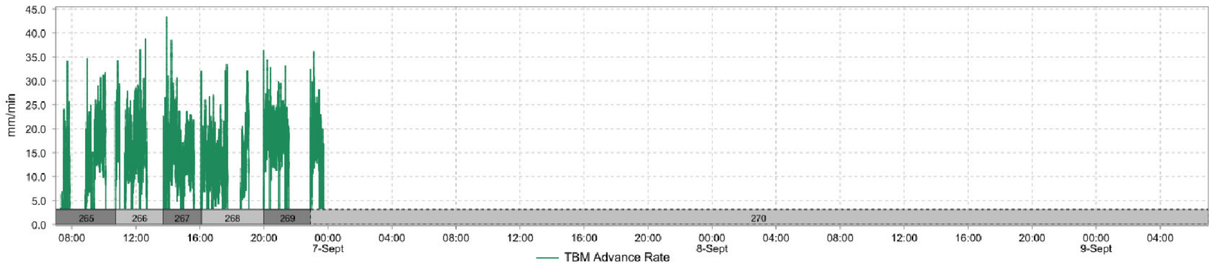

Figure 3-2 Advance rate

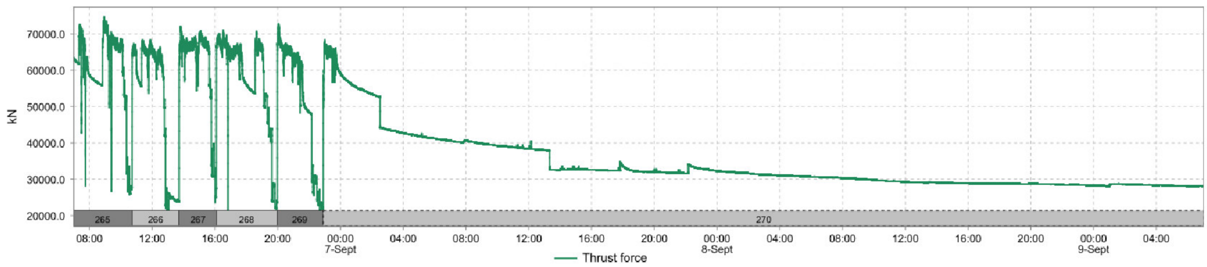

Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume closer to or above the minimum theoretical annular volume +%10 of 13.54 m^3. Ratio B/A is at 6%.

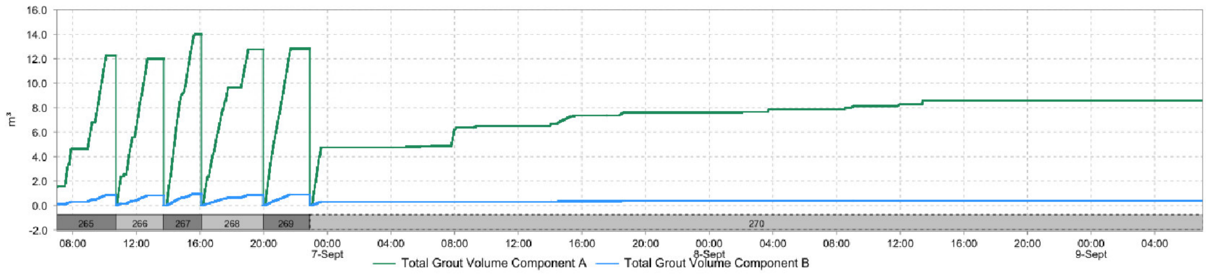

Figure 4-1 Grout volumes

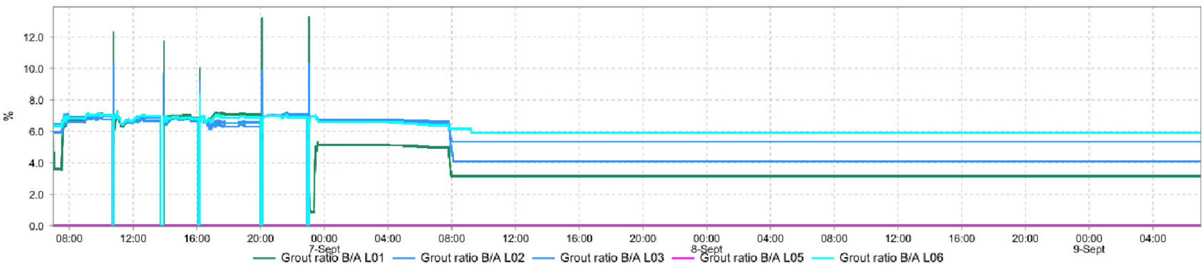

Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
265	12.3	0.8	7
266	12	0.8	7
267	14	0.9	7
268	12.8	0.9	7
269	12.8	0.9	7

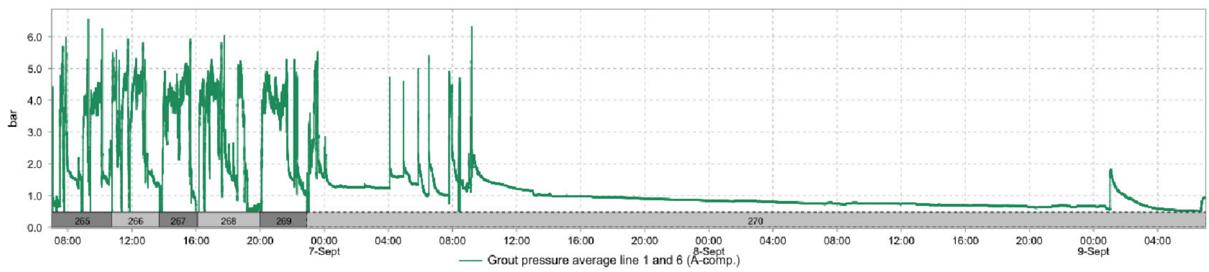

Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about -14.65 mm on the front edge of the shield and the horizontal deviation of about -19.9 mm (Figure 5-1). Maximum radial deviation is of 29.37 mm.

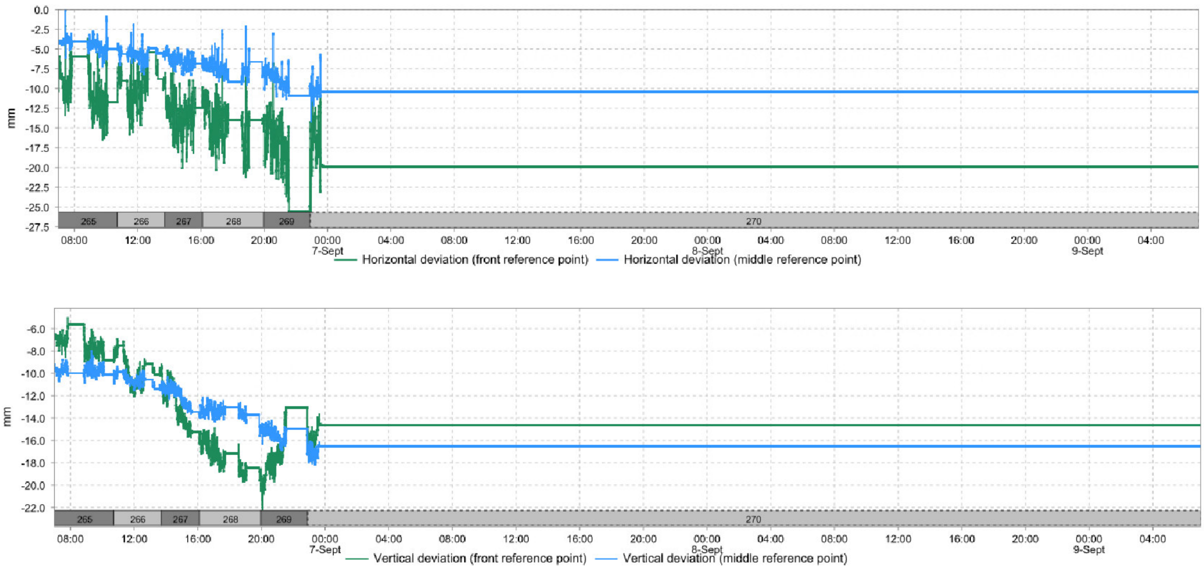

Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

Figure 6-1 Monitoring points

On PR0440, all monitoring points show downward trends starting from 06/09/2024 at 7:50 am. Points PR0440_04 and PR0440_06 have exceeded the attention threshold. Points PR0440_01, PR0440_02 and PR0440_03 have no new measures from 06/09/2024. Point PR0440_07 has exceeded the alarm threshold with a maximum settlement of 23.5 mm.

After the excavation stoppage, it is possible to observe a stabilizing trend only for points located in the ground floor, indicated with R.

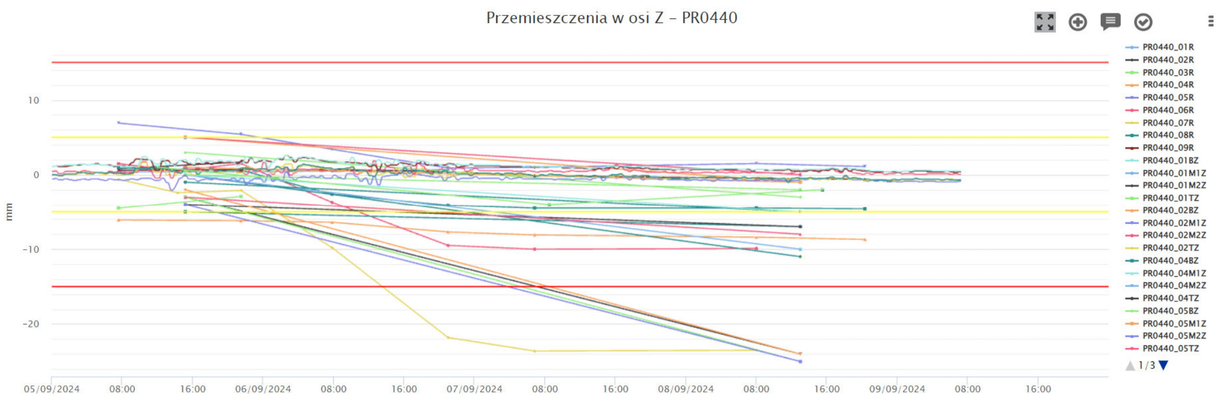

Figure 6-2 Settlement of building PR0440

On PR0460, monitoring points show a small downward trends starting from 06/09/2024 at 09:00 pm, in particular for point PR0460_07 has exceeded the attention threshold, showing then a stabilizing trend. Point PR0460_09 shows a small uplift trend.

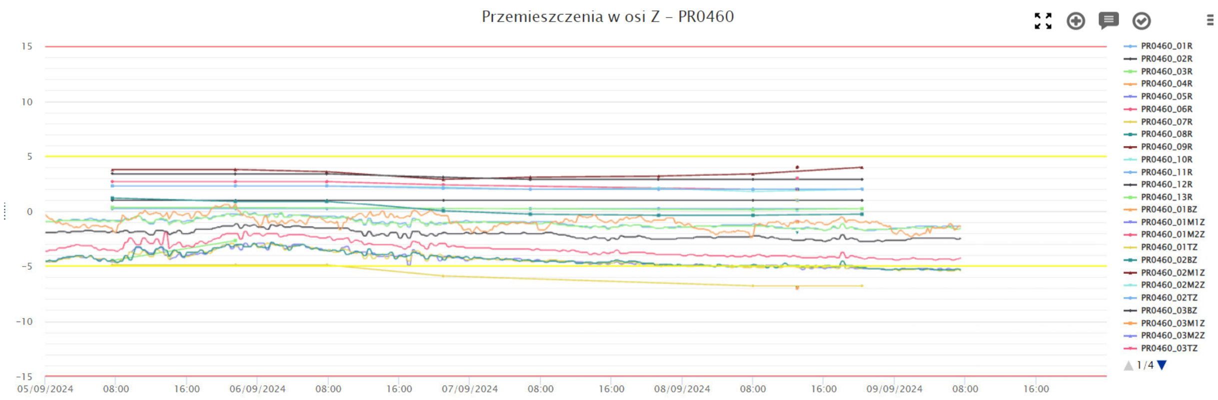

Figure 6-3 Settlement of building PR0460

On MA0190, monitoring points are stable, with small downward trends occurred during the last three days.

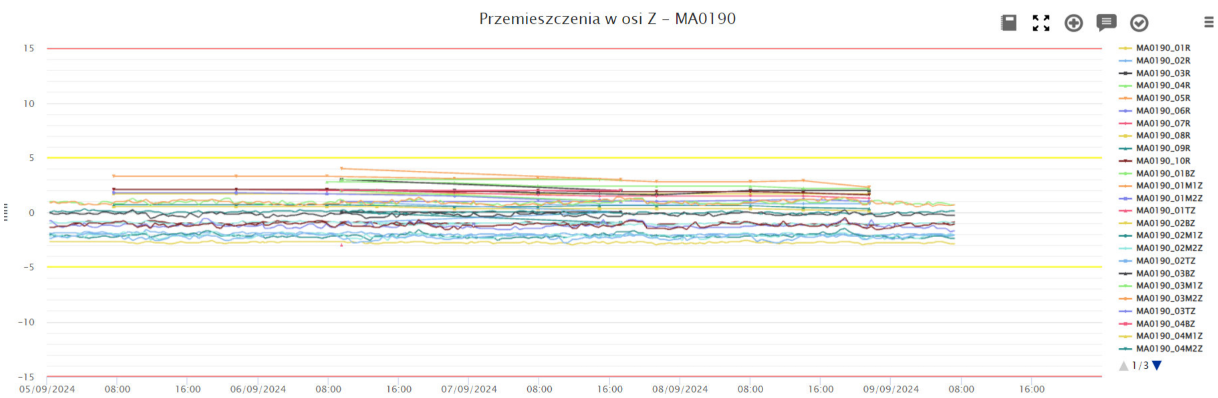

Figure 6-4 Settlement of building MA0190

On MA0210, all monitoring points show downward trends starting from 06/09/2024 at 7:50 am. Points MA0210_04 and MA0210_06 have exceeded the attention threshold. Point MA0210_05 has exceeded the alarm threshold with a maximum settlement of 24.5 mm. After the excavation stoppage, it is possible to observe a stabilizing trend.

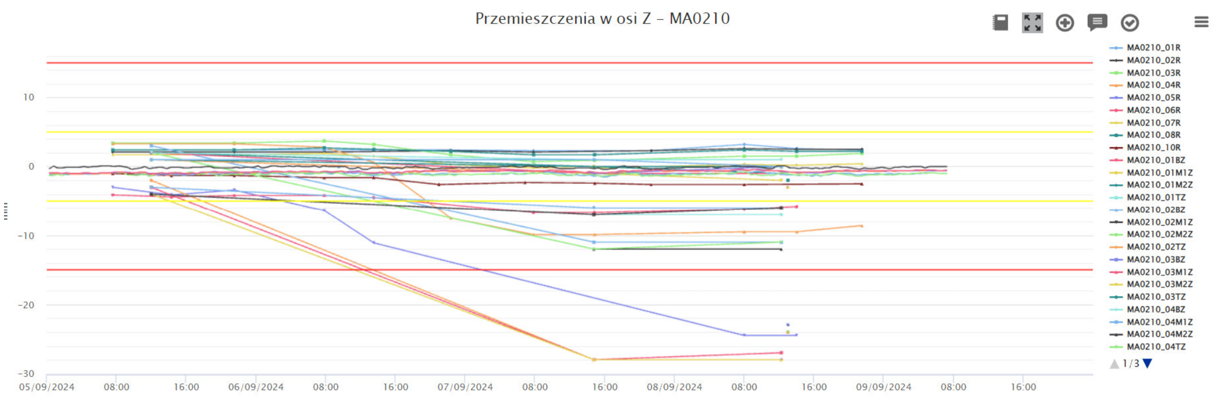

Figure 6-5 Settlement of building MA0210

On MA0230, all monitoring points show downward trends starting from 06/09/2024 at 7:50 am. Points MA0230_04 and MA0230_05 has exceeded the alarm threshold with a maximum settlement of 61 mm. After the excavation stoppage, it is possible to observe a stabilizing trend, except for MA0230_04TZ, MA0230_04BZ and MA0230_04M2Z.

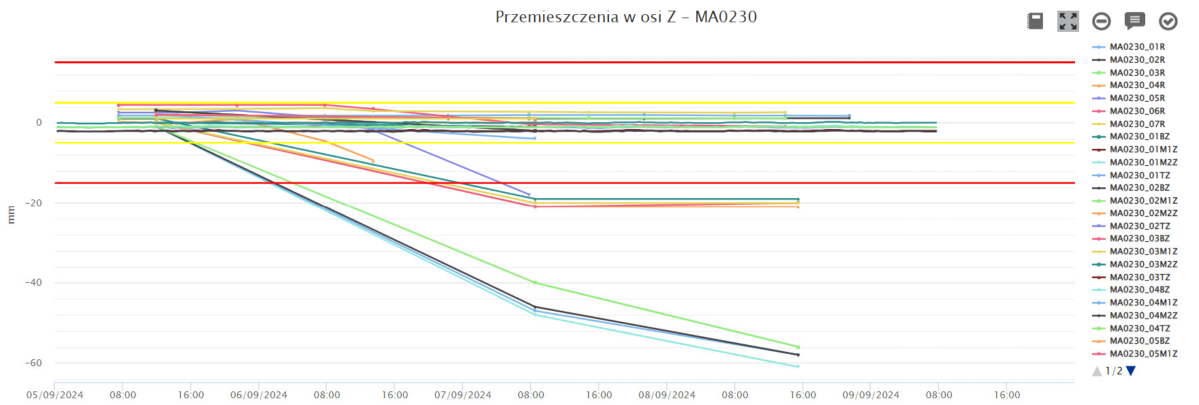

Figure 6-6 Settlement of building MA0230

On MA0250, monitoring point MA0250_01R shows an uplift trend.

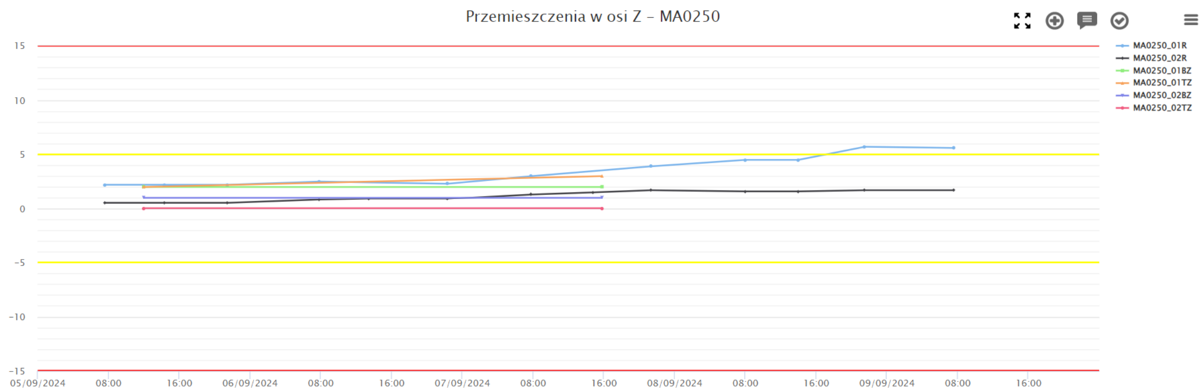

Figure 6-7 Settlement of building MA0250

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 06/09/2024 of Soletanche, compensation grouting has been performed on the 06/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 44967.12 l.

According to Raport Dobowy 07/09/2024 of Soletanche, compensation grouting has been performed on the 07/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 36352.41 l.

According to Raport Dobowy 08/09/2024 of Soletanche, compensation grouting has been performed on the 08/09/2024 from Shaft 3 in Próchnika 42.

The total amount of declared injected volume is 21704.32 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

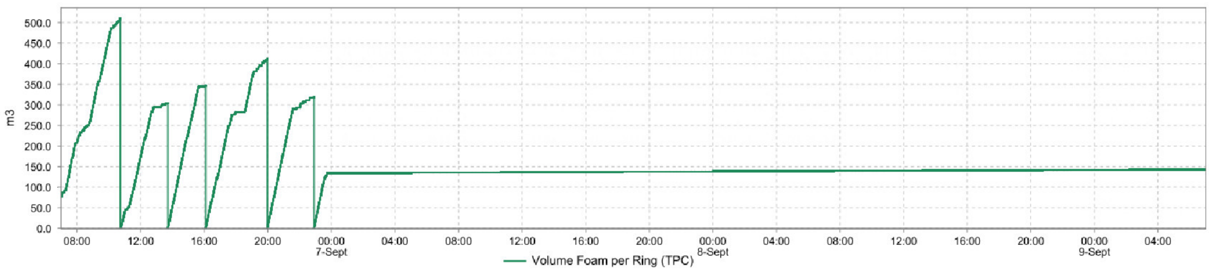

Figure 8-1 Injected foam volume

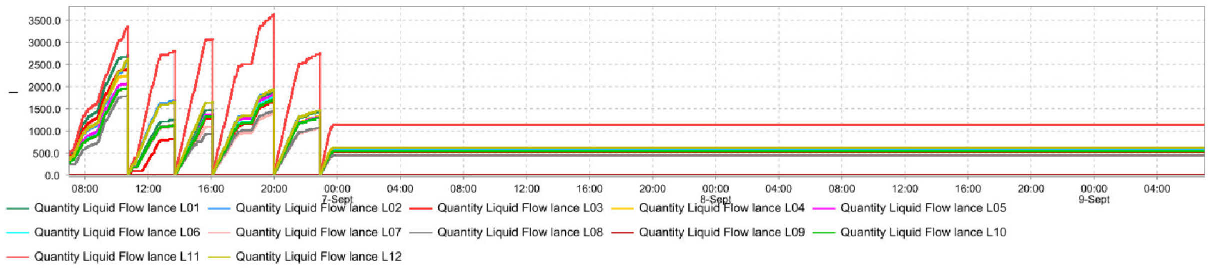

Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m^3)	C (%)	FIR (%)
265	25.7	2	241
266	14.9	2	144
267	16.5	2.1	164
268	20.9	2.2	195
269	15.8	2.1	151